

AE-378: Radiation of [Ru(CO)(OH)₂(PNP{tBu})] (AE-310) 365 nm and white, 5 mM Na₂HPO₄

Date: 2024-11-13

Tags: Dihydroxo Radiation PNP
P(tBu)N(py)P(tBu) [Ru(CO)(OH)₂(PNP)]
NMR AE 1H 31P

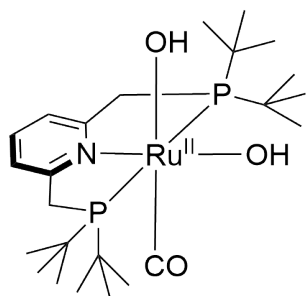
Category: Two-photon

Status: Done

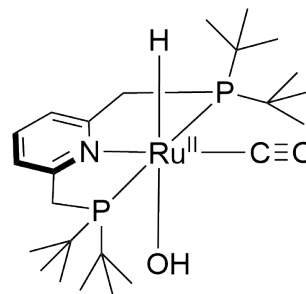
Created by: Alexander Eith

Reaction scheme/sample structure

Radiation of Dihydroxo



365 nm (0.3 W)
white (0.2 W)
H₂O



Chemical Formula: C₂₄H₄₅NO₃P₂Ru

Molecular Weight: 558,64552

[Ru(CO)(OH)₂(PNP^{tBu})]

Chemical Formula: C₂₄H₄₅NO₂P₂Ru

Molecular Weight: 542,64652

[Ru(CO)H(OH)(PNP^{tBu})]

Literature/reference experiments

Literature	Procedure after https://doi.org/10.1039/D1EE01053K
Reproduction	/
Related experiment	Two-photon - AE-376: Radiation of [Ru(CO)(OH) ₂ (PNP{tBu})] (AE-310) 365 nm and white, 5 mM Na ₂ CO ₃ Two-photon - AE-377: Radiation of [Ru(CO)(OH) ₂ (PNP{tBu})] (AE-310) 365 nm and white, 50 mM KOH in D ₂ O

Reagents

Name	CAS/Experiment number	Amount [μmol]	Equivalents	Mass _{theo} [mg]	Mass _{exp} [mg]	Molar mass [g/mol]	Volume [ml]
[Ru(CO)(OH) ₂ (PNP{tBu})]	AE-310: Synthesis of [Ru(CO)(OH) ₂ (PNP ^{tBu})] using Ag ₂ O, 5 h (AE-310-1)	4.48	/	2.50	2.54	558.65	/
Na ₂ HPO ₄	Two-photon - AE-375: Stock solutions of Na ₂ PO ₄ and Na ₂ CO ₃ in H ₂ O and KOH in D ₂ O (AE-375-1)	/	/	/	/	/	0.045

Milli-Q water	KRA-104: Degassing of Milli-Q water	/	/	/	/	/	0.85
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Irradiation Parameters

Power measured using [Power Meter] 843-R-USB + 919P-020-12 unless specified otherwise.

	Light Source Name	Wavelength [nm]	Power Setting [mW]
First light source	Light Source - LCS-0365-76-22	365 nm	300
Second light source	Light Source - LCS-6500-65-22	white	200

Used beam combiner [Name or None]	Beam Combiner - LCS-BC25-0409
Irradiation distance [cm]	9.5
Thermostat temperature [°C]	25
Irradiation duration [h]	3

Procedure/observations

All steps were carried out using [Protocol] Schlenk Technique. All steps were carried out in the dark, unless other specified.

Date	Time	Step	Observations
12.11		A 10 mL flask was prepared	
	15:05	To the flask containing the remaining [Ru] from AE-310-1 water was added using a 1 mL syringe	
		To that flask Na ₂ CO ₃ (0.1 M) was added using a 50 µL Hamilton syringe	
		The obtained mixture was transferred into the prepared 10 mL flask	
	15:12 - 15:22	The obtained mixture was stirred at 400 rpm.	before filt.jpg
	15:31	From the obtained solution an NMR sample was prepared according to Protocol - Preparation of NMR Sample using a D ₂ O/H ₃ PO ₄ capillary. (AE-378-1) During the transfer the sample was filtered through a syringe filter (PA, pore diameter = 0.2 µm)	in NMR tube.jpg

14.11		The irradiation setup from Two-photon - AE-376: Radiation of [Ru(CO)(OH)2(PNP{tBu})] (AE-310) 365 nm and white, 5 mM Na2CO3 was used without any modifications	
		The NMR-tube was placed in the setup according to [Protocol] Irradiation setup	
	13:06 - 16:09	The J.Young NMR tube was irradiated for 3:00 h	irrad setup.jpg
		An NMR was measured (AE-378-2)	AE-378-2.jpg

Analysis

Date	Time	Sample name	Analysis Method	Used Device	Solvent	Raw Data	Processed Data	Comparative Data	Interpretation
13.11	05:01	AE-378-1	31P {1H}quant, 1Hzgespg	IAAC 400 MHz I	H2O	AE-378-1.zip	AE-378-2_10.nmrium	/	As expected. Signal for Na2HPO4 can be seen. Equals approx. 0.75 equiv of [Ru]
15.11	17:26	AE-378-2	31P {1H}quant, 1Hzgespg	IAAC 400 MHz I	H2O	AE-378-2.zip	AE-378-2_10.nmrium	Two-photon - AE-376: Radiation of [Ru(CO)(OH)2(PNP{tBu})] (AE-310) 365 nm and white, 5 mM Na2CO3 (AE-376-2)	Very low conversion

Results

Sample	Yield (hydride species, cis) [%]	Yield (hydride species, trans) [%]	Yield (dihydroxo) [%]	Yield (main side product) [%]	Yield (soluble side products) [%]	Yield (other side products) [%]	Data
AE-378-2	1	0	96	0	12	- 9	AE-379-2_Yield_Determination.xlsx

Linked experiments

- [AEI-022: Radiation of \[Ru\(CO\)\(OH\)2\(PNN\)\], NO2-route, 365 nm and white LED](#)
- [AEI-026: Capillary production](#)
- [AEI-064: Synthesis of \[Ru\(CO\)\(OH\)2\(PNP\)\] with 2,6-bis\(P,P-di\(tertbutyl\)phosphinomethyl\) using Ag2O](#)
- [AEI-067: Radiation of \[Ru\(CO\)\(OH\)2\(PNP{tBu}\)\], crude, 365 nm and white](#)
- [AEI-071: Radiation of \[Ru\(CO\)\(OH\)2\(PNP{tBu}\)\], purer, 365 nm and white](#)

- AE-105: Synthesis of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP})]$ with 2,6-bis(P,P-di(tertbutyl)phosphinomethyl) using Ag_2O

- AE-110: Radiation of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$, purified, 365 nm and white

- AE-119: Radiation of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$, purified, 52 mM KOH, 365 nm and white

- AE-140: Radiation of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$, purified, 365 nm and white, longer duration I

- AE-145: Synthesis of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP})]$ with 2,6-bis(P,P-di(tertbutyl)phosphinomethyl) using Ag_2O

- AE-158: Radiation of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$, purified, 365 nm and white

- KRA-092: Radiation of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNPtBu})]$, 365 nm and white

- KRA-104: Degassing of Milli-Q water

- AE-238: Synthesis of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNPtBu})]$ using Ag_2O , 5 h

- AE-243: 1 M Stocksolution of KOH in water

- AE-246: Radiation of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$ (AE-238, 97 % pure) 365 nm and white

- AE-310: Synthesis of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNPtBu})]$ using Ag_2O , 5 h

Prep work - AE-361: 5 mM Stocksolution of KOH in water

Two-photon - AE-348: Radiation of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$ (AE-310) 365 nm and white, water, longer irradiation

Two-photon - AE-349: Radiation of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$ (AE-310) 365 nm and white, 50 mM KOH, shorter irradiation

Two-photon - AE-357: Radiation of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$ (AE-310) 365 nm and white, 1000 mM KOH

Two-photon - AE-358: Radiation of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$ (AE-310) 365 nm and white, 50 mM KOH

Two-photon - AE-362: Radiation of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$ (AE-310) 365 nm and white, 5 mM KOH

Two-photon - AE-363: Radiation of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$ (AE-310) 365 nm and white, 1 mM KOH

Two-photon - AE-371: Radiation of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$ (AE-310) 365 nm and white, 0.1 mM KOH

Two-photon - AE-375: Stock solutions of Na_2PO_4 and Na_2CO_3 in H_2O and KOH in D_2O

Two-photon - AE-376: Radiation of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$ (AE-310) 365 nm and white, 5 mM Na_2CO_3

Two-photon - AE-377: Radiation of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$ (AE-310) 365 nm and white, 50 mM KOH in

Linked resources

Beam Combiner - [Beam Combiner LCS-BC25-0409 -1](#)

Light Source - [UHP LED 365 nm-1](#)

Light Source - [UHP LED 6500 K \(white\)-1](#)

Protocol - [Preparation of NMR Sample](#)

Protocol - [Schlenk Technique](#)

Protocol - [Irradiation setup](#)

Attached files

AE-379-2_Yield_Determination.xlsx

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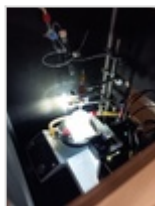
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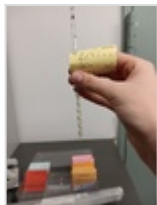
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in-NMR-tube.jpg

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before-filt.jpg

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Unique eLabID: 20241113-71aec013db103bbfc0376b80c4b0493d04bf9f22

Link: <https://elab.water-splitting.org/experiments.php?mode=view&id=1498>